Optimization 1
ISE 5406

Lecture 5

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Where We Are in the Course

Last Time (Lecture 4): Convex Functions and Characterizations

- Convex and strictly convex functions
- First-order characterization of convexity (gradient condition)
- Second-order characterization (Hessian and positive semidefiniteness)
- Operations preserving convexity
- Continuity and differentiability of convex functions

Today (Lecture 5): Hyperplanes, Subgradients, and Optimization

- Separating and supporting hyperplane theorems
- Subgradients of convex and concave functions
- Subdifferential and its properties
- Convex optimization: local implies global
- Necessary and sufficient optimality conditions

Looking Ahead

Next Time (Lecture 6): Constrained Optimization

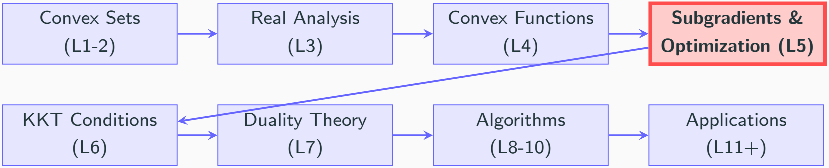
- Karush-Kuhn-Tucker (KKT) conditions
- Lagrangian duality
- Constraint qualification conditions

Coming Soon (Lecture 7): Duality Theory

- Weak and strong duality
- Dual problem formulation
- Economic interpretation of dual variables

Big Picture: Today's optimality conditions form the theoretical foundation for all constrained optimization algorithms and KKT theory.

Big Picture: Course Roadmap



Why Subgradients and Optimality Conditions Matter

- Subgradients extend the gradient to non-differentiable convex functions
- The key result: for convex problems, local optimality $\Rightarrow$ global optimality
- Optimality conditions provide the foundation for all optimization algorithms

Reading for This Lecture

Textbook References

- Sections 2.4, 3.2 and 3.4 of Bazaraa et al. (2006)
- Sections 2.5, 3.1 and 3.2 of Boyd and Vandenberghe

Key Topics

- Separating and supporting hyperplane theorems
- Subgradients and subdifferentials
- Optimality conditions for convex optimization

Outline

Roadmap

Separating and Supporting Hyperplanes

- Separating Hyperplane Theorem
- Supporting Hyperplane Theorem

Subgradients of Convex and Concave Functions

- Existence of Subgradients

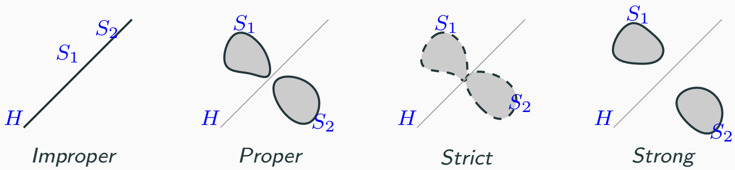
Convex Optimization

- Necessary and Sufficient Conditions for a Global Minimum

Recap: Separating Hyperplanes

Let S_1 and S_2 be nonempty sets in $\mathbb{R}^n$.

- A hyperplane $H := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T\mathbf{x} = \alpha\}$ separates S_1 and S_2 if $\mathbf{p}^T\mathbf{x} \geq \alpha$ for each $\mathbf{x} \in S_1$ and $\mathbf{p}^T\mathbf{x} \leq \alpha$ for each $\mathbf{x} \in S_2$.
- If, additionally, $S_1 \cup S_2 \not\subset H$, then H properly separates S_1 and S_2.
- The hyperplane H strictly separates S_1 and S_2 if $\mathbf{p}^T\mathbf{x} > \alpha$ for each $\mathbf{x} \in S_1$ and $\mathbf{p}^T\mathbf{x} < \alpha$ for each $\mathbf{x} \in S_2$.
- The hyperplane H strongly separates S_1 and S_2 if for *some* $\epsilon > 0$, $\mathbf{p}^T\mathbf{x} \geq \alpha + \epsilon$ for each $\mathbf{x} \in S_1$ and $\mathbf{p}^T\mathbf{x} \leq \alpha$ for each $\mathbf{x} \in S_2$.



Exercise: Types of Separation

Quick Check

For each pair of sets below, determine the **strongest** type of separation possible:

1. $S_1 = \{(x, y) : x^2 + y^2 \leq 1\}$ and $S_2 = \{(x, y) : (x - 3)^2 + y^2 \leq 1\}$
2. $S_1 = \{(x, y) : x \leq 0\}$ and $S_2 = \{(x, y) : x \geq 0\}$
3. $S_1 = \{(x, y) : x < 0\}$ and $S_2 = \{(x, y) : x > 0\}$

Hint: Consider the distance between the sets.

Solution: Types of Separation

Answers

1. Strong separation. The two unit circles centered at $(0, 0)$ and $(3, 0)$ are distance 1 apart. The hyperplane $x = 1.5$ strongly separates them with $\epsilon = 0.5$.
2. Improper separation. Both sets share the line $x = 0$, so the hyperplane $x = 0$ separates them but not properly (both sets are not entirely on one side).
3. Strict separation. The sets are disjoint but have no distance between them (they approach $x = 0$ but don't include it). The hyperplane $x = 0$ strictly separates them.

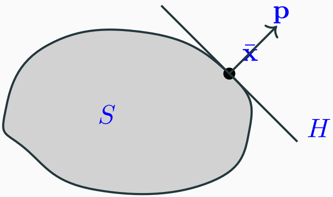
Recap: Supporting Hyperplanes

Let S be a nonempty set in $\mathbb{R}^n$ and $\bar{\mathbf{x}} \in \partial S$.

- A hyperplane $H := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) = 0\}$ is called a supporting hyperplane of S at $\bar{\mathbf{x}}$ if S is *entirely* contained in one of the halfspaces

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0\}, \quad \text{or} \quad \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0\}$$

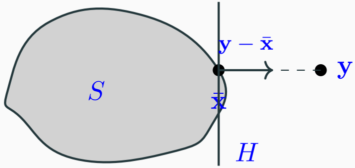
- If $S \not\subseteq H$, then H is a proper supporting hyperplane of S at $\bar{\mathbf{x}}$.



Closest-Point Theorem

Given a nonempty closed convex set $S \subseteq \mathbb{R}^n$ and a point $\mathbf{y} \in \mathbb{R}^n$ with $\mathbf{y} \notin S$, there exists a unique point $\bar{\mathbf{x}} \in S$ with minimum distance from $\mathbf{y}$. Moreover, $\bar{\mathbf{x}}$ is the minimizing point if and only if

$$(\mathbf{y} - \bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) \leq 0, \text{ for all } \mathbf{x} \in S.$$



Proof: Closest-Point Theorem

Theorem (restated)
Given nonempty closed convex S and $\mathbf{y} \notin S$, there exists a unique closest point $\bar{\mathbf{x}} \in S$, characterized by $(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ for all $\mathbf{x} \in S$.

Proof Outline: (to be done on board)

1. **Existence:** Use compactness argument (closed + bounded level sets)
2. **Uniqueness:** Suppose two closest points; derive contradiction via strict convexity of $\|\cdot\|^2$
3. **Characterization (⇒):** Use first-order optimality for $\min_{\mathbf{x} \in S} \|\mathbf{y} - \mathbf{x}\|^2$
4. **Characterization (⇐):** Show the inequality implies $\bar{\mathbf{x}}$ minimizes distance

Separating Hyperplane Theorem

Theorem (Separation of a Convex Set and a Point)
Let S be a nonempty closed convex set in $\mathbb{R}^n$ and $\mathbf{y} \notin S$. Then, there exists a nonzero vector $\mathbf{p}$ and a scalar α such that

$$\mathbf{p}^T\mathbf{y} > \alpha \text{ and } \mathbf{p}^T\mathbf{x} \leq \alpha \text{ for each } \mathbf{x} \in S.$$

Corollaries

1. Let S be a closed convex set in $\mathbb{R}^n$. Then S is the intersection of all halfspaces containing S.
2. Let S be a nonempty set, and let $\mathbf{y} \notin \text{cl}(\text{conv}(S))$. Then there exists a strongly separating hyperplane separating S and $\{\mathbf{y}\}$.

Theorems 2.4.8 and 2.4.10 in Bazaraa et al. present sufficient conditions for (strong) separation of two convex sets.

Proof: Separating Hyperplane Theorem

Theorem (restated)
Let S be nonempty closed convex and $\mathbf{y} \notin S$. Then $\exists$ nonzero $\mathbf{p}$ and α with $\mathbf{p}^T\mathbf{y} > \alpha$ and $\mathbf{p}^T\mathbf{x} \leq \alpha$ for all $\mathbf{x} \in S$.

Proof Outline: (to be done on board)

1. Apply Closest-Point Theorem to get $\bar{\mathbf{x}} \in S$ closest to $\mathbf{y}$
2. Set $\mathbf{p} = \mathbf{y} - \bar{\mathbf{x}}$ (nonzero since $\mathbf{y} \notin S$)
3. Set $\alpha = \mathbf{p}^T\bar{\mathbf{x}}$
4. Show $\mathbf{p}^T\mathbf{y} = \mathbf{p}^T\bar{\mathbf{x}} + \|\mathbf{p}\|^2 > \alpha$
5. Use characterization: $(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ implies $\mathbf{p}^T\mathbf{x} \leq \alpha$

See instructor proof notes for complete details.

Exercise: Separating Hyperplane

Group Exercise

Consider the closed convex set $S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 4\}$ (disk of radius 2 centered at origin) and the point $\mathbf{y} = (3, 0)$.

1. Find the closest point $\bar{\mathbf{x}} \in S$ to $\mathbf{y}$.
2. Find a separating hyperplane $H = \{\mathbf{x} : \mathbf{p}^T\mathbf{x} = \alpha\}$.
3. Verify that this hyperplane satisfies the separation conditions.

Hint: Use the Closest-Point Theorem. The closest point lies on the line from origin to $\mathbf{y}$*.*

Solution: Separating Hyperplane

Answers

1. Closest point: The line from origin to $\mathbf{y} = (3, 0)$ intersects the boundary of S at $\bar{\mathbf{x}} = (2, 0)$. This is the closest point since $\|\mathbf{y} - \bar{\mathbf{x}}\| = 1$.
2. Separating hyperplane: Take $\mathbf{p} = \mathbf{y} - \bar{\mathbf{x}} = (1, 0)$ and $\alpha = \mathbf{p}^T\bar{\mathbf{x}} = 2$. The hyperplane is $H = \{(x, y) : x = 2\}$.
3. Verification:
   - For $\mathbf{y}$: $\mathbf{p}^T\mathbf{y} = 3 > 2 = \alpha$ ✓
   - For any $\mathbf{x} \in S$: $x \leq 2$ by definition of S, so $\mathbf{p}^T\mathbf{x} = x \leq 2 = \alpha$ ✓

Supporting Hyperplane Theorem

Theorem (Supporting Hyperplane)
Let S be a nonempty convex set in $\mathbb{R}^n$ and $\bar{\mathbf{x}} \in \partial S$. Then there exists a hyperplane that supports S at $\bar{\mathbf{x}}$. That is, there exists $\mathbf{p} \neq \mathbf{0}$ such that

$$\mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0 \text{ for each } \mathbf{x} \in \text{cl}(S).$$

Key Insight: Every boundary point of a convex set has at least one supporting hyperplane. This is a fundamental property that connects convex geometry to optimization.

Proof: Supporting Hyperplane Theorem

Theorem (restated)
Let S be nonempty convex and $\bar{\mathbf{x}} \in \partial S$. Then $\exists$ nonzero $\mathbf{p}$ with $\mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ for all $\mathbf{x} \in \text{cl}(S)$.

Proof Outline: (to be done on board)

1. Since $\bar{\mathbf{x}} \in \partial S$, there exist points $\mathbf{y}_k \notin S$ with $\mathbf{y}_k \to \bar{\mathbf{x}}$
2. For each $\mathbf{y}_k$, apply Separating Hyperplane Theorem to get $\mathbf{p}_k$
3. Normalize: $\|\mathbf{p}_k\| = 1$ (lies on compact unit sphere)
4. Extract convergent subsequence: $\mathbf{p}_{k_j} \to \mathbf{p}$
5. Take limits in the separation inequality

See instructor proof notes for complete details.

Roadmap

Recall: Epigraph and Convexity

Theorem (from Lecture 4)
Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \to \mathbb{R}$. Then f is a convex function if and only if epi(f) is a convex set.

Key insight for today: Since epi(f) is a convex set, by the Supporting Hyperplane Theorem, every boundary point has a supporting hyperplane.

At a point $(\bar{\mathbf{x}}, f(\bar{\mathbf{x}}))$ on the graph, a supporting hyperplane to epi(f) takes the form:

$$\{(\mathbf{x}, y) : y - f(\bar{\mathbf{x}}) \geq \boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}})\}$$

for some vector $\boldsymbol{\xi}$. This vector $\boldsymbol{\xi}$ is called a subgradient.

Connection
Supporting hyperplanes to epi(f) $\longleftrightarrow$ Subgradients of f

Motivation: From First-Order Conditions to Subgradients

Recall: First-Order Characterization of Convexity (Lecture 4)
If f is **differentiable** and convex on convex S, then for all $\mathbf{x}, \bar{\mathbf{x}} \in S$:

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}})$$

The tangent line (hyperplane) at any point is a **global underestimator**.

The Problem: Many important convex functions are **not differentiable everywhere**:

- $f(x) = |x|$ (non-differentiable at $x = 0$)
- $f(\mathbf{x}) = \|\mathbf{x}\|_1 = \sum_i |x_i|$ (non-differentiable when any $x_i = 0$)
- $f(\mathbf{x}) = \max\{g_1(\mathbf{x}), \ldots, g_m(\mathbf{x})\}$ (non-differentiable at "kinks")

The Solution: Replace $\nabla f(\bar{\mathbf{x}})$ with a subgradient $\boldsymbol{\xi}$ that still provides a global underestimator!

Definition of Subgradient

Definition (Subgradient of Convex Function)
Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \to \mathbb{R}$ be a convex function. Then $\boldsymbol{\xi}$ is called a subgradient of f at $\bar{\mathbf{x}} \in S$ if

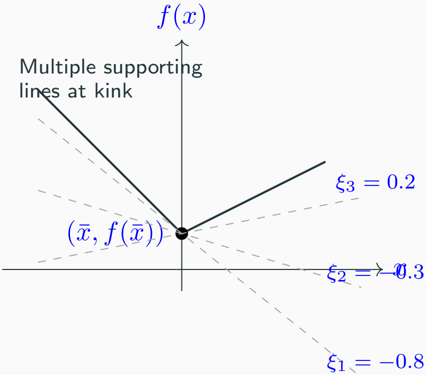
$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \quad \text{for all } \mathbf{x} \in S.$$

Key Observation: This is exactly the first-order condition, but we no longer require $\boldsymbol{\xi} = \nabla f(\bar{\mathbf{x}})$. At points where f is differentiable, the gradient is the unique subgradient.

Definition (Subgradient of Concave Function)
Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \to \mathbb{R}$ be a concave function. Then $\boldsymbol{\xi}$ is called a subgradient of f at $\bar{\mathbf{x}} \in S$ if

$$f(\mathbf{x}) \leq f(\bar{\mathbf{x}}) + \boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \quad \text{for all } \mathbf{x} \in S.$$

Geometric Interpretation of Subgradients



- The affine function $f(\bar{\mathbf{x}}) + \boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}})$ corresponds to a supporting hyperplane of the epigraph or hypograph depending on whether f is convex or concave.
- The subgradient vector $\boldsymbol{\xi}$ corresponds to the slope of the supporting

Exercise: Computing Subgradients

Quick Check

Consider the convex function $f(x) = |x|$ on $\mathbb{R}$.

1. What is the subgradient of f at $x = 2$?
2. What is the subgradient of f at $x = -3$?
3. What are all subgradients of f at $x = 0$?

Hint: At differentiable points, the subgradient equals the derivative. At $x = 0$, find all slopes ξ such that $|x| \geq 0 + \xi \cdot x$ for all x.

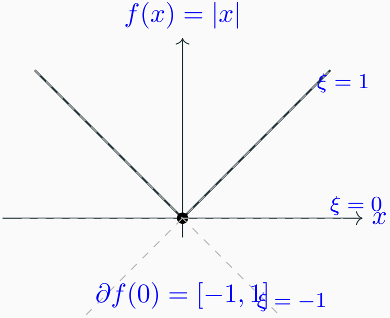
Solution: Computing Subgradients

Answers

1. At $x = 2$: f is differentiable with $f'(2) = 1$. The unique subgradient is $\xi = 1$.
2. At $x = -3$: f is differentiable with $f'(-3) = -1$. The unique subgradient is $\xi = -1$.
3. At $x = 0$: We need $|x| \geq \xi \cdot x$ for all x.
    - For $x > 0$: $x \geq \xi x \Rightarrow \xi \leq 1$
    - For $x < 0$: $-x \geq \xi x \Rightarrow \xi \geq -1$

    Thus, the subdifferential at $x = 0$ is $\partial f(0) = [-1, 1]$.

Example: Subgradients of $f(x) = |x|$



At $x = 0$, any line with slope $\xi \in [-1, 1]$ passes through the origin and stays below the graph of f, making each such ξ a valid subgradient.

Important Non-Differentiable Convex Functions

These functions arise frequently in optimization and machine learning:

1. ℓ_1 Norm (LASSO, Compressed Sensing)
$f(\mathbf{x}) = \|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|$

Non-differentiable when any $x_i = 0$. The subdifferential at $\mathbf{x}$:

$$\partial f(\mathbf{x}) = \{\boldsymbol{\xi} : \xi_i = \mathsf{sign}(x_i) \text{ if } x_i \neq 0,\ \xi_i \in [-1, 1] \text{ if } x_i = 0\}$$

2. Maximum of Affine Functions
$f(\mathbf{x}) = \max_{i=1,\ldots,m}\{\mathbf{a}_i^T\mathbf{x} + b_i\}$

Non-differentiable where multiple functions achieve the maximum. At $\mathbf{x}$:

$$\partial f(\mathbf{x}) = \mathsf{conv}\{\mathbf{a}_i : i \in I(\mathbf{x})\} \quad \text{where } I(\mathbf{x}) = \{i : \mathbf{a}_i^T\mathbf{x} + b_i = f(\mathbf{x})\}$$

Important Non-Differentiable Convex Functions (cont.)

3. Hinge Loss (Support Vector Machines)
$f(x) = \max\{0, 1 - x\}$

Non-differentiable at $x = 1$. Subdifferential:

$$\partial f(x) = \begin{cases} \{0\} & x > 1 \\ [-1, 0] & x = 1 \\ \{-1\} & x < 1 \end{cases}$$

4. ReLU (Neural Networks)
$f(x) = \max\{0, x\}$

Non-differentiable at $x = 0$. Subdifferential:

$$\partial f(x) = \begin{cases} \{1\} & x > 0 \\ [0, 1] & x = 0 \\ \{0\} & x < 0 \end{cases}$$

Existence of Subgradients

Theorem (Existence of Subgradients)
Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \to \mathbb{R}$ be a convex function. Then for $\bar{\mathbf{x}} \in \text{int}(S)$, there exists a vector $\boldsymbol{\xi}$ such that the hyperplane

$$H = \{(\mathbf{x}, y) : y = f(\bar{\mathbf{x}}) + \boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}})\}$$

supports $\text{epi}(f)$ at $[\bar{\mathbf{x}}, f(\bar{\mathbf{x}})]$. In particular, $\boldsymbol{\xi}$ is a subgradient of f at $\bar{\mathbf{x}}$.

A convex function has a subgradient at every point in the interior of its domain.

Proof: reading assignment (see Theorem 3.2.5 of Bazaraa et al.)

- If f is differentiable at $\bar{\mathbf{x}}$, then $\nabla f(\bar{\mathbf{x}})$ is the *only* subgradient of f at $\bar{\mathbf{x}}$.
- Set of all subgradients of f at $\bar{\mathbf{x}}$ is called the subdifferential $\partial f(\bar{\mathbf{x}}) \subseteq \mathbb{R}^n$.
- $\partial f(\bar{\mathbf{x}})$ is a compact convex set at interior points $\bar{\mathbf{x}}$.

Characterization via Subgradients

Theorem (Subgradient Characterization of Convexity)
Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \to \mathbb{R}$. Suppose for each point $\bar{\mathbf{x}} \in \text{int}(S)$, there exists a subgradient vector $\boldsymbol{\xi}$ such that

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \quad \text{for each } \mathbf{x} \in S.$$

Then f is convex on $\text{int}(S)$.

Proof outline:

Take any $\mathbf{x}_1, \mathbf{x}_2 \in \text{int}(S)$ and $\lambda \in [0, 1]$. Let $\bar{\mathbf{x}} = \lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2$.

By the subgradient inequality applied at $\bar{\mathbf{x}}$:

- $f(\mathbf{x}_1) \geq f(\bar{\mathbf{x}}) + \boldsymbol{\xi}^T(\mathbf{x}_1 - \bar{\mathbf{x}})$
- $f(\mathbf{x}_2) \geq f(\bar{\mathbf{x}}) + \boldsymbol{\xi}^T(\mathbf{x}_2 - \bar{\mathbf{x}})$

Multiply by λ and $(1 - \lambda)$ respectively and add to get:

Exercise: Subdifferential Computation

Group Exercise

Consider the convex function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = \max\{x_1, x_2\}$$

1. Find $\partial f(1, 0)$ (the subdifferential at $(1, 0)$).
2. Find $\partial f(0, 0)$ (the subdifferential at the origin).
3. For what points (x_1, x_2) is f differentiable?

Hint: Consider where $x_1 > x_2$, $x_1 < x_2$, and $x_1 = x_2$.

Solution: Subdifferential Computation

Answers

1. At $(1, 0)$: Since $x_1 = 1 > 0 = x_2$, we have $f(x) = x_1$ in a neighborhood. Thus f is differentiable with $\nabla f = (1, 0)^T$, so $\partial f(1, 0) = \{(1, 0)^T\}$.
2. At $(0, 0)$: Here $x_1 = x_2 = 0$. The subdifferential is the convex hull of the gradients of the constituent functions:

   $$\partial f(0, 0) = \text{conv}\{(1, 0)^T, (0, 1)^T\} = \{(\lambda, 1 - \lambda)^T : \lambda \in [0, 1]\}$$

   This is the line segment from $(1, 0)$ to $(0, 1)$.
3. Differentiable points: f is differentiable wherever $x_1 \neq x_2$. Along the line $x_1 = x_2$, there are multiple subgradients.

Roadmap

Recall: The General Optimization Problem

General Nonlinear Programming Problem (from Lecture 1)

Key difficulties with general NLP:

- Local minima $\neq$ global minima (non-convexity)
- Verifying global optimality is computationally hard (NP-hard in general)
- Optimization algorithms may converge to local minima

Today's focus: What happens when we impose convexity?

Minimizing a Convex Function

Definition (Convex Optimization Problem)
Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function and S be a nonempty convex set in $\mathbb{R}^n$. Then a convex optimization problem is defined by

$$ \tag{1} $$

Theorem (Local Implies Global)
Suppose that $\bar{\mathbf{x}} \in S$ is a local optimal solution to Problem (1).

1. Then $\bar{\mathbf{x}}$ is a global optimal solution.
2. If either $\bar{\mathbf{x}}$ is a strict local minimum or f is strictly convex, then $\bar{\mathbf{x}}$ is the unique global optimal solution (it is also a strong local minimum).

Proof: Local Implies Global

Theorem (restated)
For a convex optimization problem, any local minimum is a global minimum.

Proof:

Let $\bar{\mathbf{x}}$ be a local minimum: $\exists \epsilon > 0$ such that $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in S$ with $\|\mathbf{x} - \bar{\mathbf{x}}\| < \epsilon$.

Goal: Show $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in S$.

Take any $\mathbf{x} \in S$. If $\|\mathbf{x} - \bar{\mathbf{x}}\| < \epsilon$, done by local optimality.

Otherwise, choose $\lambda = \frac{\epsilon}{2\|\mathbf{x}-\bar{\mathbf{x}}\|} \in (0, 1)$ and let $\mathbf{y} = (1-\lambda)\bar{\mathbf{x}} + \lambda\mathbf{x}$.

Since $\|\mathbf{y} - \bar{\mathbf{x}}\| = \frac{\epsilon}{2} < \epsilon$: $\quad f(\bar{\mathbf{x}}) \leq f(\mathbf{y})$

By convexity: $f(\mathbf{y}) \leq (1-\lambda)f(\bar{\mathbf{x}}) + \lambda f(\mathbf{x})$

Combining: $f(\bar{\mathbf{x}}) \leq (1-\lambda)f(\bar{\mathbf{x}}) + \lambda f(\mathbf{x})$

Exercise: Local Implies Global

Proof Exercise

Prove that for a convex optimization problem, any local minimum is also a global minimum.

Setup: Let $f : S \to \mathbb{R}$ be convex on convex set S. Suppose $\bar{\mathbf{x}}$ is a local minimum, i.e., there exists $\epsilon > 0$ such that $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in S$ with $\|\mathbf{x} - \bar{\mathbf{x}}\| < \epsilon$.

Goal: Show $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in S$.

Hint: For any $\mathbf{x} \in S$, consider points on the line segment between $\bar{\mathbf{x}}$ and $\mathbf{x}$ that are within the ϵ-neighborhood.

Solution: Local Implies Global

Proof

Suppose $\bar{\mathbf{x}}$ is a local minimum. Take any $\mathbf{x} \in S$.

If $\|\mathbf{x} - \bar{\mathbf{x}}\| < \epsilon$, then $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ by local optimality.

If $\|\mathbf{x} - \bar{\mathbf{x}}\| \geq \epsilon$, choose $\lambda = \frac{\epsilon}{2\|\mathbf{x}-\bar{\mathbf{x}}\|} \in (0, 1)$ and let

$$\mathbf{y} = (1 - \lambda)\bar{\mathbf{x}} + \lambda\mathbf{x}$$

Then $\|\mathbf{y} - \bar{\mathbf{x}}\| = \lambda\|\mathbf{x} - \bar{\mathbf{x}}\| = \frac{\epsilon}{2} < \epsilon$, so $f(\bar{\mathbf{x}}) \leq f(\mathbf{y})$.

By convexity:

$$f(\mathbf{y}) \leq (1 - \lambda)f(\bar{\mathbf{x}}) + \lambda f(\mathbf{x})$$

Combining: $f(\bar{\mathbf{x}}) \leq (1 - \lambda)f(\bar{\mathbf{x}}) + \lambda f(\mathbf{x})$

Rearranging: $\lambda f(\bar{\mathbf{x}}) \leq \lambda f(\mathbf{x})$

Since $\lambda > 0$: $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$. $\square$

Necessary and Sufficient Conditions for a Global Minimum

Theorem (Optimality Conditions)
Point $\bar{\mathbf{x}} \in S$ is an optimal solution to Problem (1) if and only if f has a subgradient $\boldsymbol{\xi}$ at $\bar{\mathbf{x}}$ such that $\boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all $\mathbf{x} \in S$.

Corollary 1: Under the assumptions of the above theorem, if S is an open set (for example, $S = \mathbb{R}^n$), $\bar{\mathbf{x}}$ is an optimal solution to the problem if and only if there exists a zero subgradient of f at $\bar{\mathbf{x}}$.

Corollary 2: Assume that f is differentiable. Then $\bar{\mathbf{x}}$ is an optimal solution if and only if $\nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all $\mathbf{x} \in S$. Furthermore, if S is open, $\bar{\mathbf{x}}$ is an optimal solution if and only if $\nabla f(\bar{\mathbf{x}}) = \mathbf{0}$.

Proof: Optimality Conditions

Theorem (restated)
$\bar{\mathbf{x}} \in S$ is optimal iff $\exists$ subgradient $\boldsymbol{\xi}$ at $\bar{\mathbf{x}}$ with $\boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all $\mathbf{x} \in S$.

Proof ($\Rightarrow$): (Optimal $\Rightarrow$ condition holds)

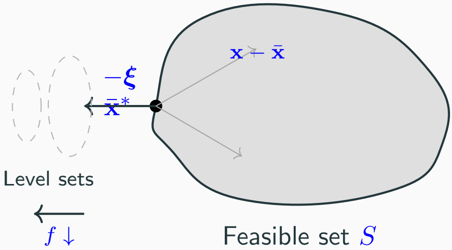
1. If $\bar{\mathbf{x}} \in \text{int}(S)$: subgradient exists; optimality $\Rightarrow \mathbf{0} \in \partial f(\bar{\mathbf{x}})$
2. If $\bar{\mathbf{x}} \in \partial S$: use epigraph and supporting hyperplane argument

Proof ($\Leftarrow$): (Condition holds $\Rightarrow$ optimal)

1. By subgradient definition: $f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}})$
2. Given condition: $\boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all $\mathbf{x} \in S$
3. Therefore: $f(\mathbf{x}) \geq f(\bar{\mathbf{x}})$ for all $\mathbf{x} \in S$ ✓

See instructor proof notes for complete details.

Geometric Interpretation of Optimality



The condition $\boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ means the negative subgradient $-\boldsymbol{\xi}$ (descent direction) points away from the feasible set.

Exercise: Optimality Conditions

Application Exercise

Consider minimizing $f(x) = x^2$ over $S = [1, 3]$.

1. What is the optimal solution $\bar{x}$?
2. Compute $\nabla f(\bar{x})$.
3. Verify the optimality condition: $\nabla f(\bar{x})^T (x - \bar{x}) \geq 0$ for all $x \in [1, 3]$.

Hint: The unconstrained minimum of f is at $x = 0$, which is not in S.

Solution: Optimality Conditions

Answers

1. Optimal solution: Since $f(x) = x^2$ is increasing for $x > 0$ and $S = [1, 3]$, the minimum occurs at $\bar{x} = 1$.
2. Gradient: $\nabla f(x) = 2x$, so $\nabla f(\bar{x}) = \nabla f(1) = 2$.
3. Verification: For all $x \in [1, 3]$:

$$\nabla f(\bar{x})^T (x - \bar{x}) = 2(x - 1) \geq 0$$

   since $x \geq 1$ for all $x \in S$. ✓

   The gradient points in the direction of increasing x, but all feasible directions from $\bar{x} = 1$ also go in that direction, confirming $\bar{x} = 1$ is optimal.

Exercise: End-of-Lecture Assessment

True or False

Determine whether each statement is true or false:

1. Every convex function has a subgradient at every point in its domain.
2. If f is convex and differentiable, then $\partial f(\mathbf{x}) = \{\nabla f(\mathbf{x})\}$.
3. A local minimum of a convex function over a convex set is always a global minimum.
4. The subdifferential $\partial f(\mathbf{x})$ is always a convex set.
5. Strong separation implies strict separation.

Solution: End-of-Lecture Assessment

Answers

1. False. A convex function has a subgradient at every point in the *interior* of its domain, but not necessarily at boundary points.
2. True. For differentiable convex functions, the gradient is the unique subgradient.
3. True. This is the fundamental theorem of convex optimization.
4. True. The subdifferential is always a closed convex set (in fact, compact at interior points).
5. True. Strong separation (with gap $\epsilon > 0$) implies the sets don't touch, which is the condition for strict separation.

Summary and Next Steps

Key Takeaways

- **Separating hyperplanes:** Convex sets can be separated from external points; types include proper, strict, and strong separation
- **Supporting hyperplanes:** Every boundary point of a convex set has a supporting hyperplane
- **Subgradients:** Generalize gradients to non-differentiable convex functions; subdifferential $\partial f(\mathbf{x})$ is always convex
- **Local $\Rightarrow$ Global:** For convex optimization, any local minimum is a global minimum
- **Optimality conditions:** $\bar{\mathbf{x}}$ is optimal iff $\exists \boldsymbol{\xi} \in \partial f(\bar{\mathbf{x}})$ with $\boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all feasible $\mathbf{x}$

Next Time:

- Karush-Kuhn-Tucker (KKT) conditions for constrained optimization
- Lagrangian duality and its applications

Questions?